

This tasks goal is to Analyze website traffic logs to understand user journeys, top landing pages, bounce rates and referral sources.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('ggplot')
```

1. loading data

```
df = pd.read_csv('traffic.csv')

df.head()
```

	event	date	country	city	artist	album	track	isrc	linkid	
0	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8	
1	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8	
2	click	2021-08-21	India	Ludhiana	Reyanna Maria	So Pretty	So Pretty	USUM72100871	23199824-9cf5-4b98-942a-34965c3b0cc2	
3	click	2021-08-21	France	Unknown	Simone & Simaria, Sebastian Yatra	No Llores Más	No Llores Más	BRUM72003904	35573248-4e49-47c7-af80-08a960fa74cd	
4	click	2021-	Meldives	Melá	Tesher	Jalebi	Jalebi	QZNWQ2070741	2d896d31-97b6-4869-	

```
print(df.shape)
```

```
print(df.info())
```

```
print(df.isnull().sum())
```

```
(226278, 9)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 226278 entries, 0 to 226277
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0   event       226278 non-null object
1   date        226278 non-null object
2   country     226267 non-null object
3   city        226267 non-null object
4   artist      226241 non-null object
5   album       226273 non-null object
6   track       226273 non-null object
7   isrc        219157 non-null object
8   linkid      226278 non-null object
dtypes: object(9)
memory usage: 15.5+ MB
None
event      0
date       0
```

```
country      11
city         11
artist       37
album        5
track        5
isrc         7121
linkid       0
dtype: int64
```

```
df.drop_duplicates(inplace=True)
```

```
df['date'] = pd.to_datetime(df['date'])
```

```
df.fillna('Unknown', inplace=True)
```

```
#total events
total_events = len(df)

print("Total Events:", total_events)

#countries
print("Countries:", df['country'].nunique())

#cities
print("Cities:", df['city'].nunique())

#Artists
print("Artists:", df['artist'].nunique())

#tracks
print("Tracks:", df['track'].nunique())
```

```
Total Events: 122567
Countries: 212
Cities: 11993
Artists: 2420
Tracks: 3563
```

2.event distribution and visualization

```
event_counts = df['event'].value_counts()

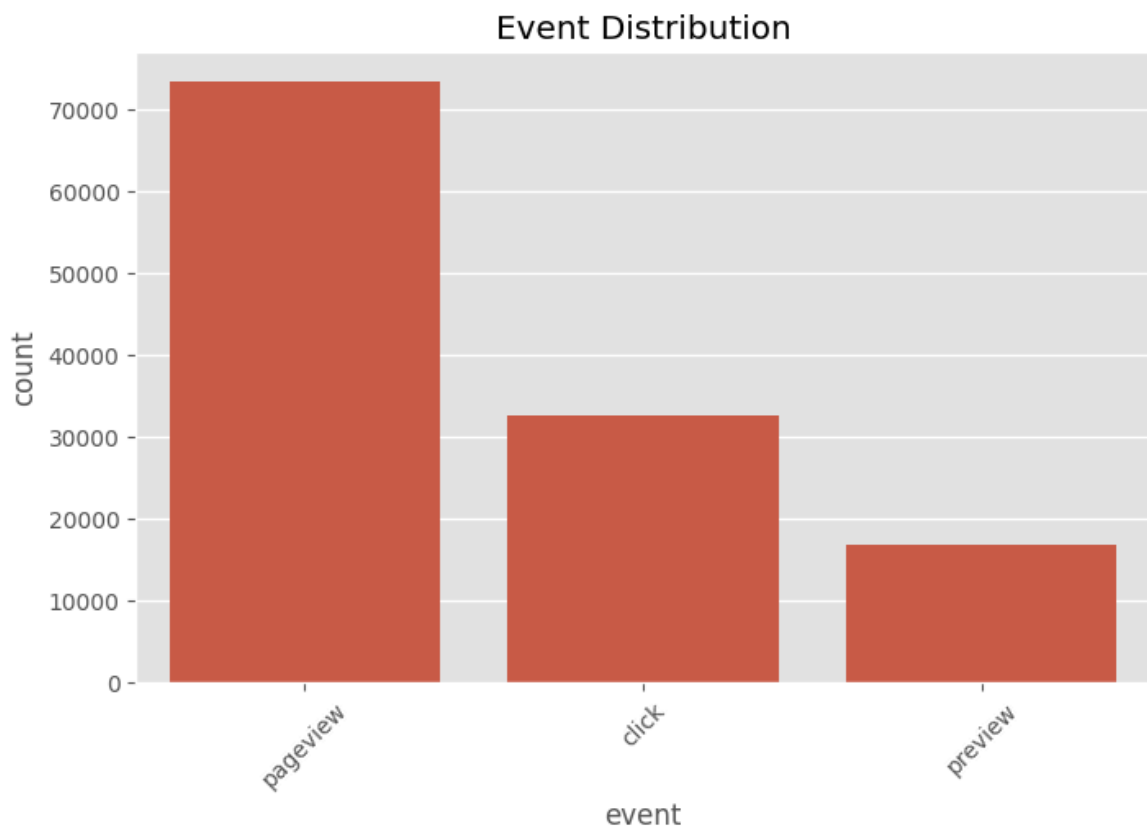
print(event_counts)
```

```
event
pageview    73360
click       32499
preview     16708
Name: count, dtype: int64
```

```
#visualization
plt.figure(figsize=(8,5))

sns.countplot(
    x='event',
    data=df,
    order=df['event'].value_counts().index
)

plt.title("Event Distribution")
plt.xticks(rotation=45)
plt.show()
```



3.top contries and visualization

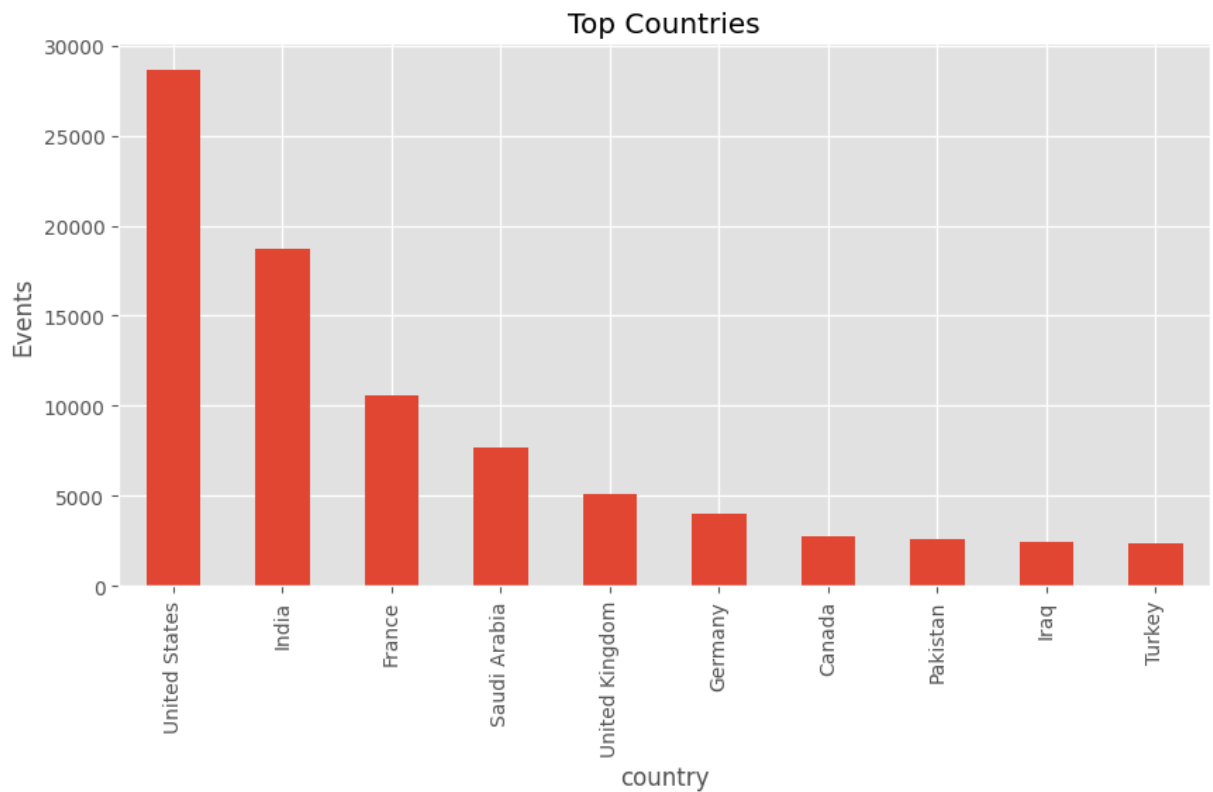
```
top_countries = df['country'].value_counts().head(10)

print(top_countries)
```

```
country
United States    28664
India            18689
France           10565
Saudi Arabia      7682
United Kingdom    5095
Germany           4017
Canada            2784
Pakistan          2633
Iraq              2444
Turkey            2399
Name: count, dtype: int64
```

```
#visualization
top_countries.plot(
    kind='bar',
    figsize=(10,5)
)

plt.title("Top Countries")
plt.ylabel("Events")
plt.show()
```



4. top cities and visualization

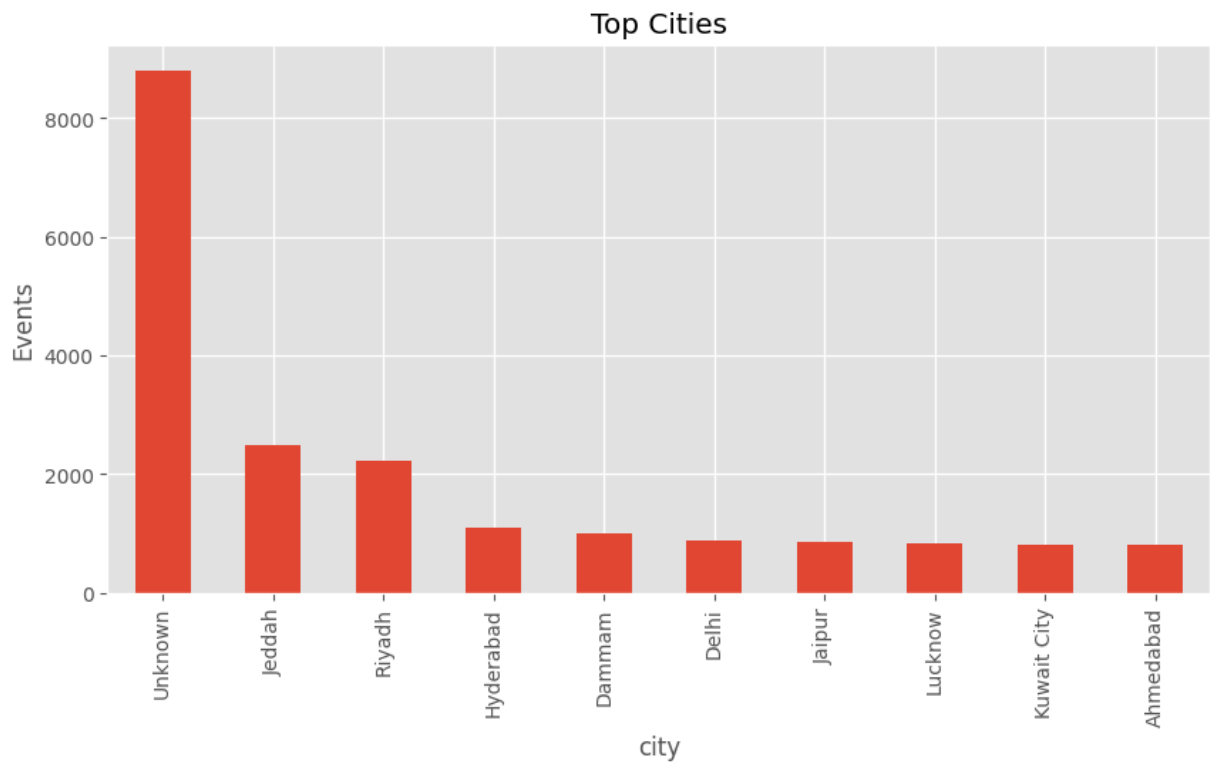
```
top_cities = df['city'].value_counts().head(10)

print(top_cities)
```

```
city
Unknown      8802
Jeddah       2497
Riyadh       2232
Hyderabad    1088
Dammam       1002
Delhi        884
Jaipur       849
Lucknow      837
Kuwait City  816
Ahmedabad    808
Name: count, dtype: int64
```

```
#visualization
top_cities.plot(
    kind='bar',
    figsize=(10,5)
)

plt.title("Top Cities")
plt.ylabel("Events")
plt.show()
```



5. popular artists and visualization

```
top_artists = df['artist'].value_counts().head(10)
```

```
print(top_artists)
```

```

artist
Teshar                                8288
Anne-Marie                            4029
Tundra Beats                          3951
Roddy Ricch                           3107
Olivia Rodrigo                        3037
Surf Mesa, Emilee                     2956
DMNDS, Strange Fruits Music, Fallen Roses, Lujavo, Nito-Onna  2865
Reyanna Maria                         2675
PinkPantheress                        2446
50 Cent, Olivia                       2390
Name: count, dtype: int64

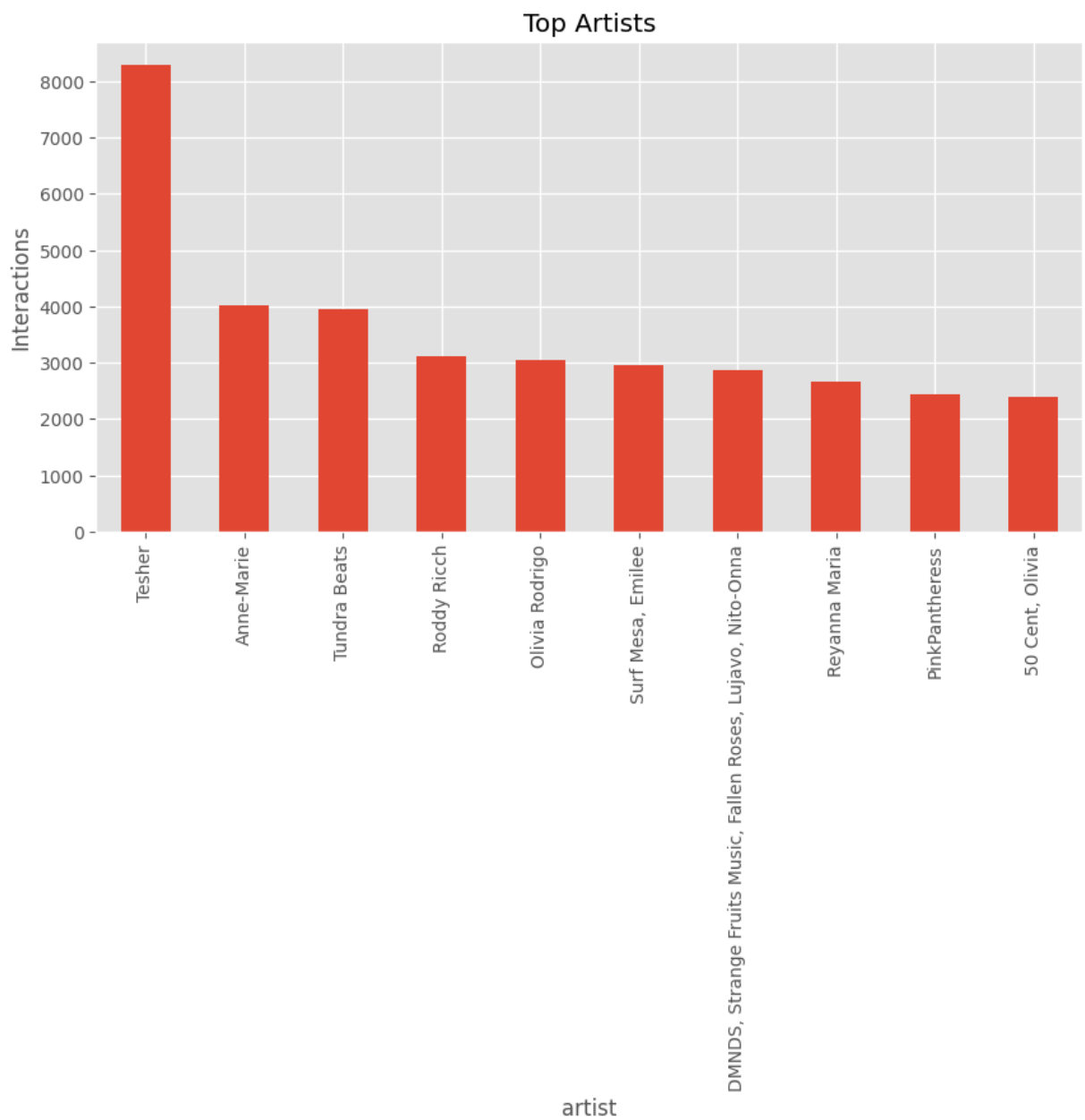
```

```

#visualization
top_artists.plot(
    kind='bar',
    figsize=(10,5)
)

plt.title("Top Artists")
plt.ylabel("Interactions")
plt.show()

```



6. popular albums and visualization

```
top_albums = df['album'].value_counts().head(10)

print(top_albums)
```

```
album
Jalebi Baby                8288
Beautiful                  4028
Beautiful Day              3950
Late At Night              3059
ily (i love you baby) (feat. Emilee)  2956
SOUR                       2878
The Massacre               2870
Calabria (feat. Lujavo & Nito-Onna)  2865
So Pretty                  2830
Summer Of Love             2108
Name: count, dtype: int64
```

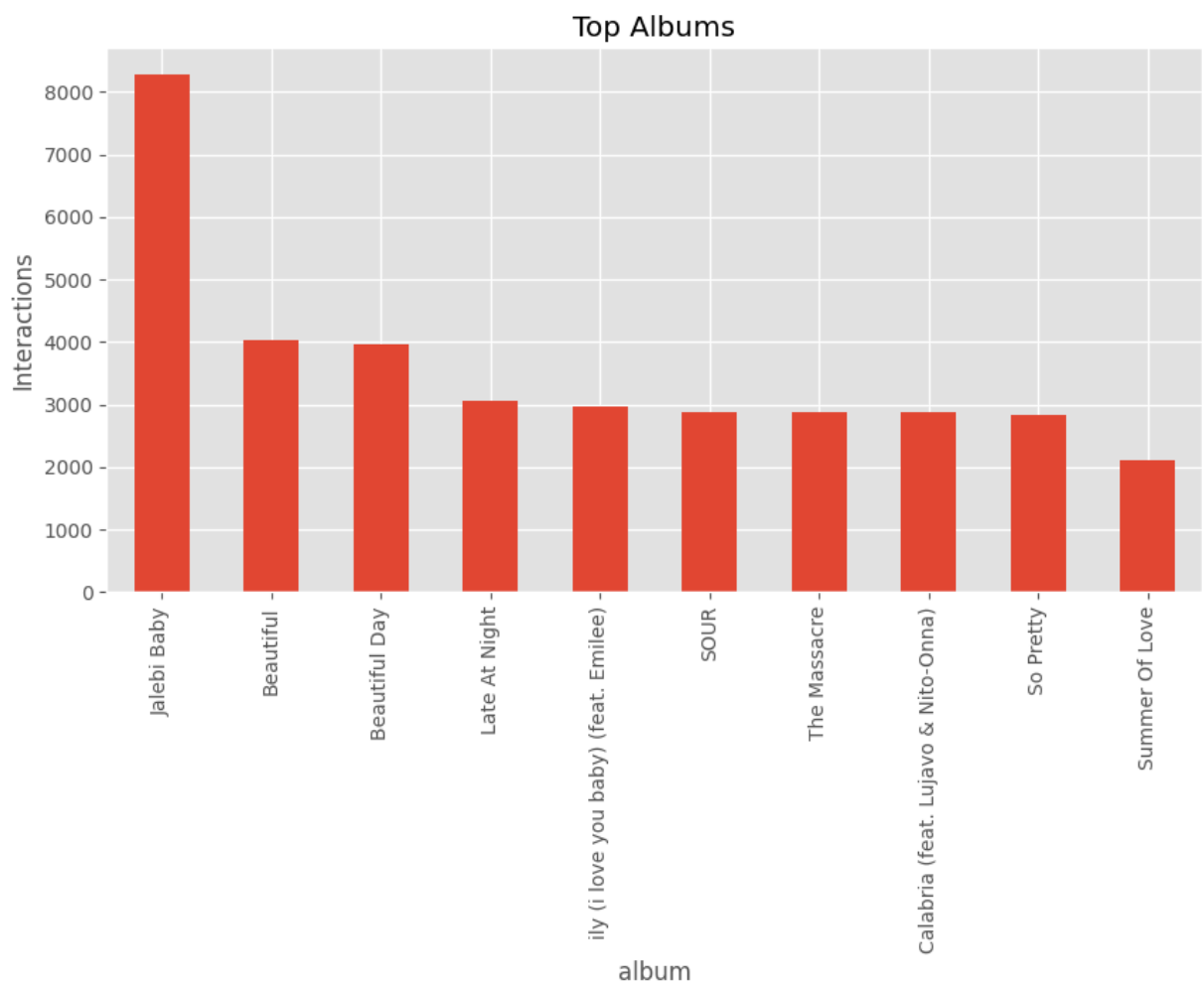
```
#visualization
top_albums.plot()
```

```

kind='bar',
figsize=(10,5)
)

plt.title("Top Albums")
plt.ylabel("Interactions")
plt.show()

```



7. popular tracks and visualization

```

top_tracks = df['track'].value_counts().head(10)

print(top_tracks)

```

```

track
Jalebi Baby                8288
Beautiful                  4037
Beautiful Day              3951
Late At Night              3059
ily (i love you baby) (feat. Emilee)  2956
Calabria (feat. Lujavo & Nito-Onna)  2865
So Pretty                  2830
Candy Shop                 2397
Summer of Love (Shawn Mendes & Tainy)  2108
Build a Bitch              2072
Name: count, dtype: int64

```

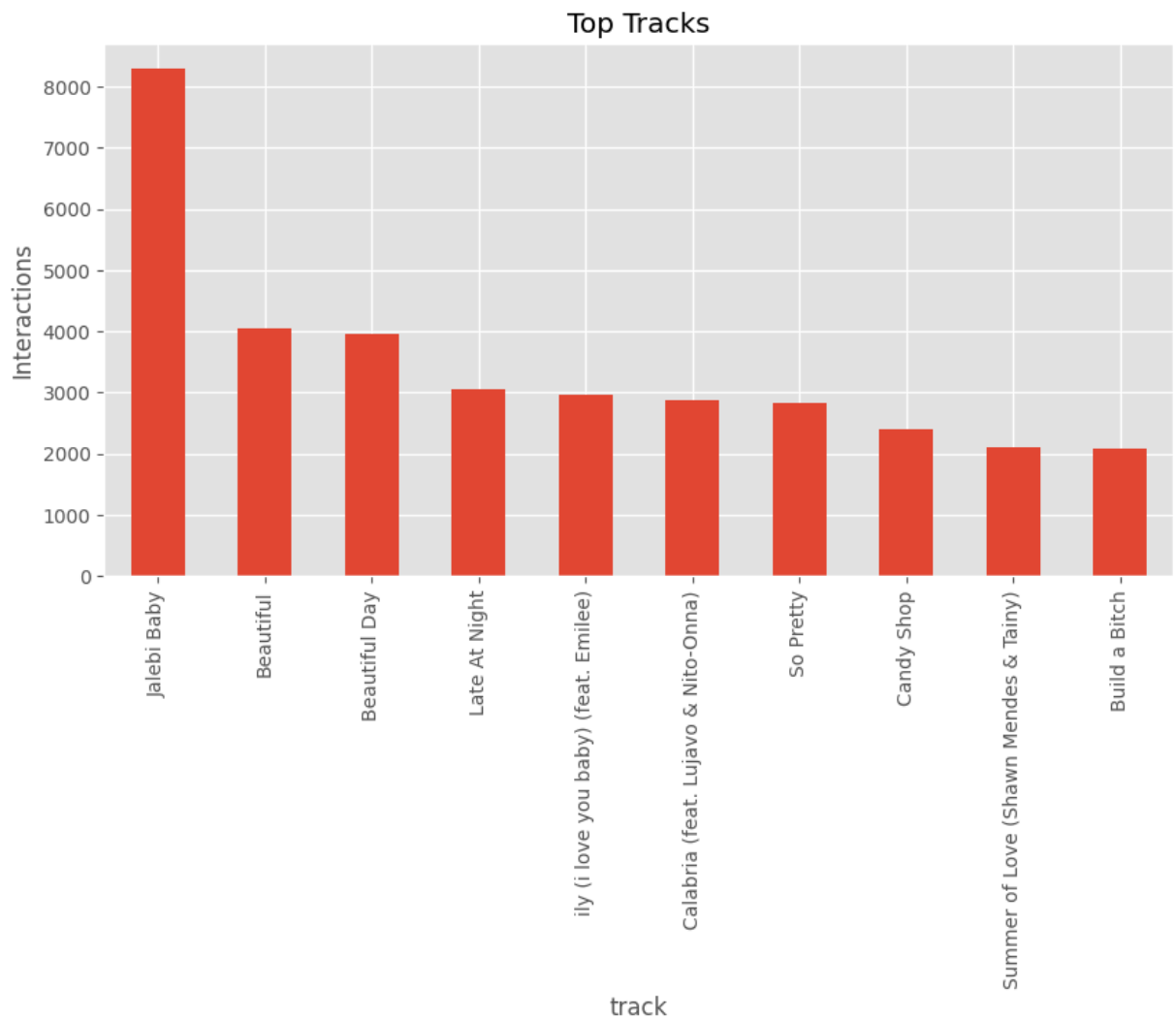
```

#visualization
top_tracks.plot(
    kind='bar',
    figsize=(10,5)
)

```

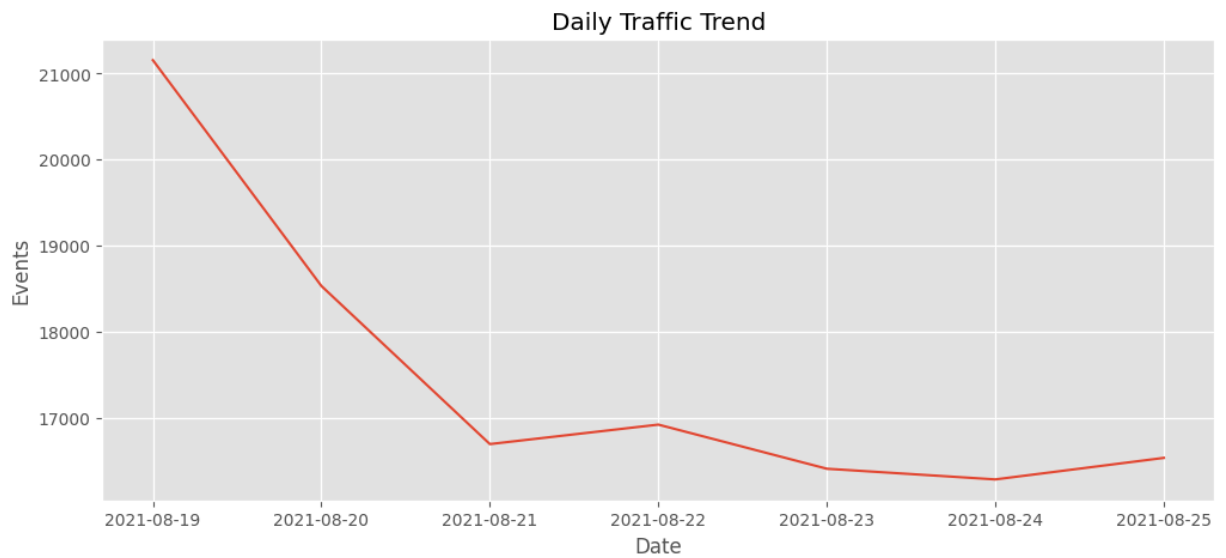
```
)
```

```
plt.title("Top Tracks")  
plt.ylabel("Interactions")  
plt.show()
```



```
daily_traffic = df.groupby(  
    df['date'].dt.date  
) .size()
```

```
plt.figure(figsize=(12,5))  
  
daily_traffic.plot()  
  
plt.title("Daily Traffic Trend")  
plt.xlabel("Date")  
plt.ylabel("Events")  
plt.show()
```

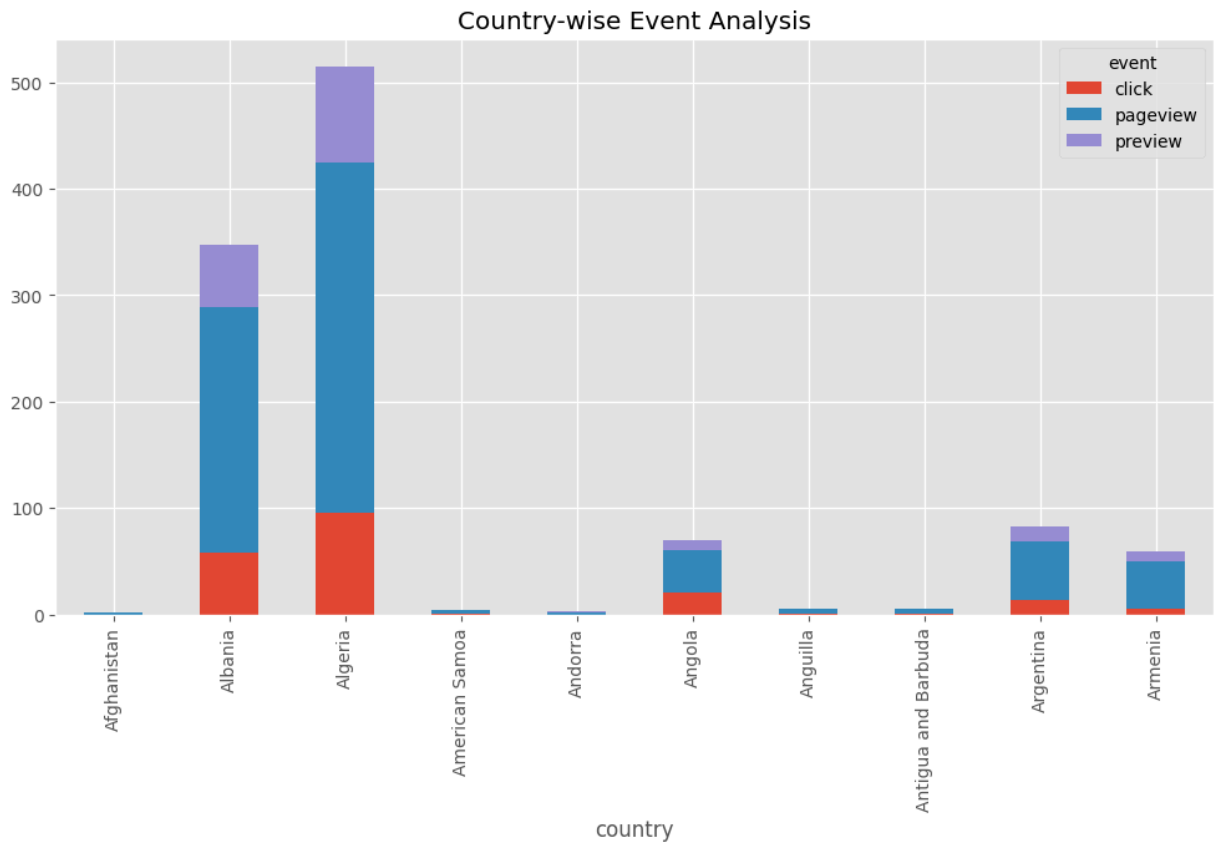
8. country wise event analysis

```
country_event = pd.crosstab(  
    df['country'],  
    df['event']  
)  
  
country_event.head()
```

	event	click	pageview	preview
country				
Afghanistan		0	2	0
Albania		58	231	58
Algeria		96	328	91
American Samoa		1	3	0
Andorra		0	2	1

Next steps: [Generate code with country_event](#) [New interactive sheet](#)

```
#visualization  
country_event.head(10).plot(  
    kind='bar',  
    stacked=True,  
    figsize=(12,6)  
)  
  
plt.title("Country-wise Event Analysis")  
plt.show()
```



```
#artist engagement analysis
artist_event = pd.crosstab(
    df['artist'],
    df['event']
)
```

```
artist_event.head(10)
```

	event	click	pageview	preview
artist				
\$Mar		0	1	0
\$NOT		3	5	1
\$hyfromdatre		1	1	0
\$krtr Cobain		0	1	0
\$ofaygo		2	5	0
\$uicideboy\$		2	6	0
*GLITCH		0	1	0
03 Greedo		0	2	0
1 King		1	3	0
187 Strassenbande		0	1	0

Next steps:

[Generate code with artist_event](#)

[New interactive sheet](#)

top link id's

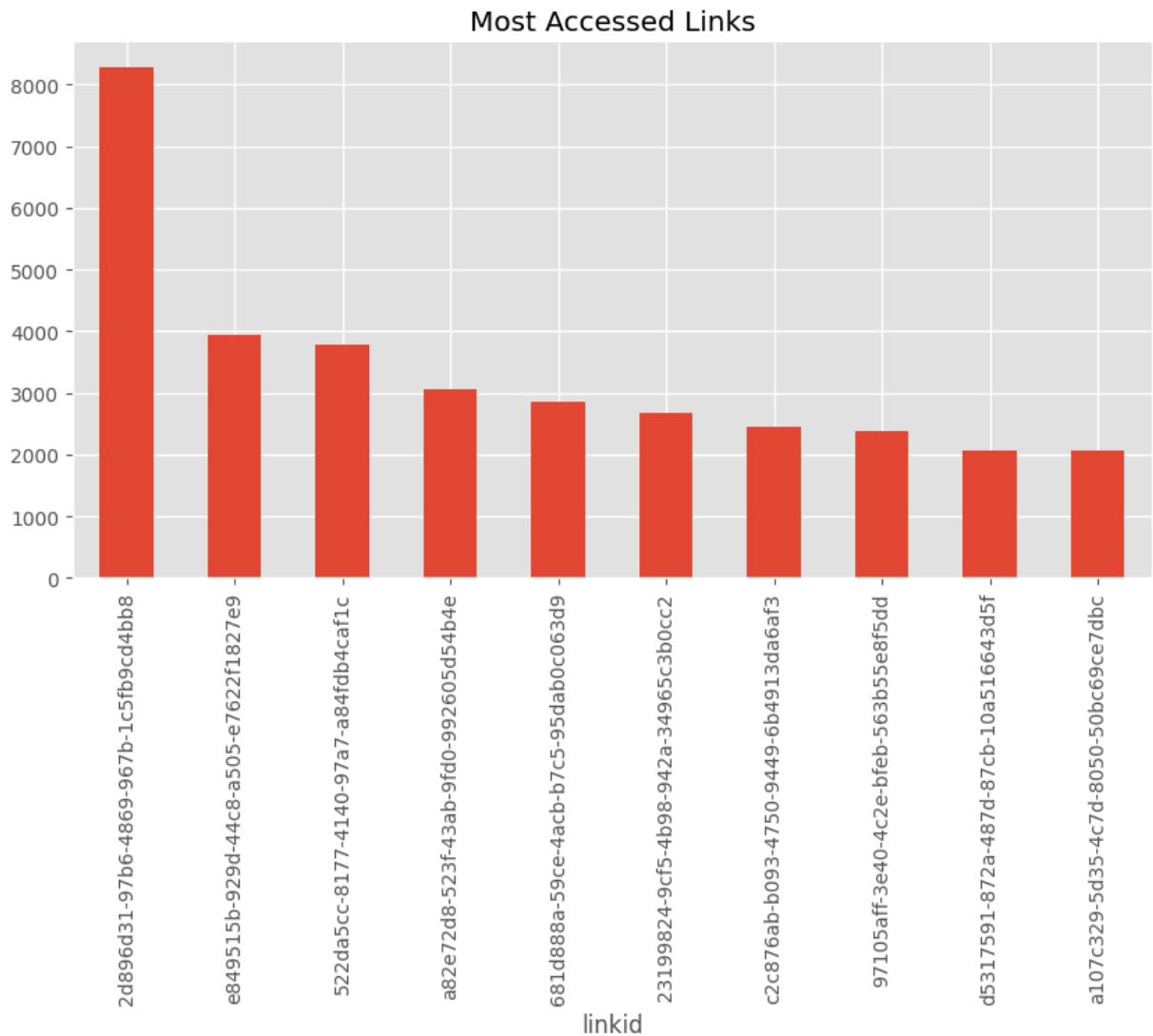
```
top_links = df['linkid'].value_counts().head(10)
```

```
print(top_links)
```

```
linkid
2d896d31-97b6-4869-967b-1c5fb9cd4bb8    8288
e849515b-929d-44c8-a505-e7622f1827e9    3950
522da5cc-8177-4140-97a7-a84fdb4caf1c    3786
a82e72d8-523f-43ab-9fd0-992605d54b4e    3059
681d888a-59ce-4acb-b7c5-95dab0c063d9    2865
23199824-9cf5-4b98-942a-34965c3b0cc2    2675
c2c876ab-b093-4750-9449-6b4913da6af3    2442
97105aff-3e40-4c2e-bfeb-563b55e8f5dd    2390
d5317591-872a-487d-87cb-10a516643d5f    2074
a107c329-5d35-4c7d-8050-50bc69ce7dbc    2072
Name: count, dtype: int64
```

```
#visualization
top_links.plot(
    kind='bar',
    figsize=(10,5)
)

plt.title("Most Accessed Links")
plt.show()
```



RECOMMENDATIONS

1. Focus on High-Traffic Regions